On the unitary dual of the metaplectic cover of SL(2, F), F p-adic

A corrected and annotated revision of the 1999 manuscript

D. Joyner (with `Argon-Gemini` and `Sol-ChatGPT`)

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Abstract

This AI-aided paper is a corrected version of a paper originally written in 1999. Argon-Gemini was used to referee, suggest corrections, and add a glossary of terms. Sol-ChatGPT was used as a second referee. See the acknowledgments at the end of the paper for details.

This revision revisits a 20-something year old unpublished manuscript of mine concerning the n-fold metaplectic cover of $\mathrm{SL}(2, F)$. It gives an explicit and corrected description of the genuine representations of the covering diagonal torus, establishes the vanishing and image statements for normalized split orbital integrals, and states the principal-series character formula in terms of genuine Weyl conjugation, rather than inversion in the covering group. For $\gcd(p, 2n) = 1$ and n odd or $4 \mid n$, the resulting formula yields a precisely normalized split-torus character transfer to $\mathrm{SL}(2, F)$. The remaining discussion surveys principal series and known complementary-series examples, distinguishing results established here from additional assertions requiring separate references or proof.

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Table of notation

The principal notation is collected here for reference. Definitions and normalization conventions are given in Section 3.

Symbol	Meaning
$F, \mathcal{O}_F, \varpi, q$	The local field; its ring of integers; a uniformizer; and the residue-field cardinality.
$n,\ d = n/\gcd(n,2)$	Covering degree n and the dimension d of a genuine irreducible representation of the covering torus.
$G, \widetilde{G}$	$\mathrm{SL}(2,F)$ and its n-fold Kubota cover.
$A, \widetilde{A}$	The diagonal torus and its full inverse image under the covering map.
Z_A	Center of the covering torus $\widetilde{A}$; explicitly $\rho^{-1}(h((F^\times)^d))$.
$\widetilde{C}$	A maximal abelian subgroup of $\widetilde{A}$, lying above $h(\varpi^{d\mathbb{Z}}\mathcal{O}_F^\times)$.
λ, χ_λ	A genuine central character of Z_A, and the associated degree-d representation of $\widetilde{A}$.
$W, \tilde{w}$	Weyl group and the chosen lift of its nontrivial element.
$\Phi_f,\ \mathcal{O}_f^{Z_A}$	Normalized split orbital integrals, respectively with quotient by $\widetilde{A}$ on Z_A and by Z_A on $\widetilde{A}$.
D, δ_B	Weyl discriminant and modulus character of the Borel subgroup B.
$\mathrm{I}(\chi)$	Normalized principal series induced from $\chi\delta_B^{1/2}$.
ν, η_λ	Power-section homomorphism (for the permitted degrees) and the associated pulled-back character.

1 Introduction

Let F be a nonarchimedean local field of characteristic zero and residue characteristic p, and let

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{G} \xrightarrow{\rho} G = \mathrm{SL}(2, F) \longrightarrow 1$$

be Kubota's n-fold covering group. We work with representations on which μ_n acts by a fixed faithful character. In the metaplectic setting, even the preimage of the diagonal torus A need not be abelian. This simple phenomenon is responsible both for the higher dimension of its genuine irreducible representations and for the vanishing of their characters away from the radical of the commutator pairing.

The main objectives of this corrected note are:

(i) to describe the genuine representation theory of $\widetilde{A}$ by a finite Heisenberg-group argument (Theorem 4.3);

(ii) to formulate normalized split orbital integrals using an actual centralizing subgroup and to identify their image (Theorem 5.2);

(iii) to give the principal-series character on the split regular set with its correct central equivariance (Theorem 7.1);

(iv) to deduce a character comparison for the degrees in which the naive power-section map is genuinely multiplicative (Theorem 8.3).

The last result is a *split-torus/principal-series* comparison; no comparison of elliptic orbital integrals or all genuine admissible representations is asserted.

The historical antecedents, relevant representation-theoretic results, and subsequent developments are described in the literature review in Section 2. A principal distinction is between results about individual principal series representations and results that classify the entire genuine unitary dual.

Assumptions. The statements about the radical and Stone–von Neumann theory below hold whenever $|\mu_n(F)| = n$. We explicitly assume $\gcd(p, 2n) = 1$ when using a maximal compact splitting, the tame parametrization of $\widetilde{A}$, or the image theorem. The power-map transfer in Section 8 has the additional restriction n odd or $4 \mid n$; the omitted congruence class is discussed in Remark 8.5.

2 Literature review

This review is selective. It emphasizes the results directly used or compared with the covering-torus calculations, orbital-integral theory, principal series, and representation-theoretic transfers in this note. In particular, a result for an n-fold cover of $\mathrm{GL}(2, F)$ need not immediately imply an analogous statement for the cover of $\mathrm{SL}(2, F)$.

The covering cocycle and the metaplectic representation. Kubota [K] constructs the covering group of $\mathrm{SL}(2, F)$ by an explicit two-cocycle expressed in terms of the Hilbert symbol; this is the starting point for the commutator calculation in Section 4. Gelbart and Sally [GS] investigate intertwining operators and the Weil representation for the metaplectic group; Gelbart's monograph [G] develops its representation and spectral theory. These works explain why exceptional constituents of induced representations must be considered separately from generic principal series.

Principal series, intertwining operators, and reducibility. Moen's 1979 thesis [Mo1] studies intertwining operators for covering groups of $\mathrm{SL}(2)$, while her later article [Mo2] proves reducibility and irreducibility results for the *unitary* principal series (under its stated hypotheses, including odd residue characteristic). In particular, the behavior depends on the parity of n. Aritürk [Ar] determines the composition series at the exceptional parameters in a family of genuine principal series of the *threefold* cover; the special parameters are not a classification of complementary series in arbitrary degree. The general theory of parabolic induction and Jacquet modules for linear reductive p-adic groups is developed in Bernstein–Zelevinsky [BZ1, BZ2]; applying these techniques to a cover requires attention to genuine inducing data.

Correspondence and the larger metaplectic setting. Flicker [F] studies automorphic forms on covering groups of $\mathrm{GL}(2)$. Kazhdan and Patterson [KP] develop metaplectic forms for higher-rank general linear groups, while Flicker–Kazhdan [FK] provides a metaplectic correspondence framework. These papers supply a broader context for character identities but do not, by themselves, establish the complete local transfer asserted in some earlier versions of this manuscript.

Supercuspidal representations and the author's related work. Manderscheid [M1, M2] treats supercuspidal representations of $\mathrm{SL}(2)$ and its double cover. Joyner [J1] establishes a metaplectic analogue of a Kazhdan-style lifting, including local orbital-integral identities in a different setting. The generalized Hecke-algebra correspondence of [J2] applies to admissible representations of *sufficiently large level*, rather than giving an unrestricted classification. In [J3], Joyner studies supercuspidal representations of the n-fold cover in the tame setting and gives a character comparison for the double cover. The later paper [J4] discusses invariant distributions and the tempered dual, and explicitly cites the unpublished 1999 manuscript. The split orbital-integral statements revisited here are closely related to that later work.

Plancherel measures and more recent local coefficients. Goldberg and Szpruch [GoldSz] compute Plancherel measures and a matrix-valued analogue of Shahidi local coefficients for genuine principal series of covers of $\mathrm{SL}(2, F)$, obtaining irreducibility results. Szpruch [Szpruch19] studies the associated local coefficients matrix and derives further Plancherel formulas, without assuming the residue characteristic is prime to n but with a restriction that $4 \nmid n$. Tang [Tang] studies principal series of metaplectic groups more generally, including constructions of genuine central characters for the covering torus. These

later results offer checks on reducibility and on the use of intertwining operators; they should not be conflated with the limited split-character comparison established in Section 8.

3 Covering groups and conventions

Write $\mathcal{O}_F$ for the ring of integers, ϖ for a uniformizer, $q = |\mathcal{O}_F/\varpi\mathcal{O}_F|$, $|\varpi| = q^{-1}$, and $K = \mathrm{SL}(2, \mathcal{O}_F)$. We suppose that F contains μ_n of order n and identify it with a subgroup of $\mathbb{C}^\times$. All representations and functions called *genuine* use this fixed central character. Let

$$h(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, \qquad u(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus $A = h(F^\times)$, $N = u(F)$, $B = AN$, and $W = N_G(A)/A = \{1, w\}$. Write $\widetilde{A} = \rho^{-1}(A)$, $\widetilde{B} = \rho^{-1}(B)$, and use tildes for elements of the cover. The cover splits canonically over N. When $\gcd(p, 2n) = 1$ we also fix a splitting of K (chosen consistently with the cocycle); in the spherical case K henceforth denotes its image in $\widetilde{G}$.

For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ put $x(g) = c$ if $c \neq 0$ and $x(g) = d$ otherwise. We use Kubota's cocycle

$$\beta(g_1, g_2) = \left(\frac{x(g_1 g_2)}{x(g_1)}, \frac{x(g_1 g_2)}{x(g_2)} \right)_n, \tag{3.1}$$

where $(\ ,\ )_n$ is the nth Hilbert symbol. The law on $G \times \mu_n$ is $(g, \zeta)(g', \zeta') = (gg', \zeta\zeta'\beta(g, g'))$. We will write $\tilde{h}(a) = (h(a), 1)$, and $\tilde{w} = (w, 1)$. A representation π is genuine if $\pi(\zeta g) = \zeta\pi(g)$; a test function f is genuine if

$$f(\zeta g) = \zeta^{-1} f(g). \tag{3.2}$$

Hence the Harish–Chandra character of a genuine admissible representation satisfies $\Theta_\pi(\zeta g) = \zeta\Theta_\pi(g)$ on the regular set. This opposite convention for test functions is essential in trace integrals.

For $a = h(t) \in A_{\mathrm{reg}}$, where $t^2 \neq 1$, let

$$D(a) = \det\bigl(1 - \mathrm{Ad}(a)\,|_{\mathfrak{g}/\mathfrak{a}}\bigr) = (1 - t^2)(1 - t^{-2}) = -(t - t^{-1})^2, \tag{3.3}$$

$$\delta_B(a) = |t|^2, \qquad |D(a)|^{1/2} = |t - t^{-1}|. \tag{3.4}$$

These quantities pull back through ρ. Set

$$d = n_0 = \frac{n}{\gcd(n, 2)}. \tag{3.5}$$

We give μ_n Haar mass one and normalize Haar measures by $\mathrm{vol}(A \cap K) = \mathrm{vol}(N \cap K) = \mathrm{vol}(K) = 1$. Haar measure on G and $\widetilde{G}$ is chosen compatibly with the Iwasawa decomposition $G = ANK$ and the chosen splitting of K (when it exists), and quotient measures are induced from these choices. We use the *restriction* of Haar measure on A to its open subgroups; we do not silently replace it by a pushforward measure. All orbital-integral and character identities below use this common normalization.

4 The covering torus

4.1 The commutator and its radical

For $a, b \in F^\times$ an immediate calculation from (3.1) gives, up to inversion according to the commutator convention,

$$[\tilde{h}(a), \tilde{h}(b)] = (a, b)_n^{\pm 2}. \tag{4.1}$$

The sign has no effect on the following radical calculation. Define

$$Z_A = Z(\widetilde{A}) = \rho^{-1}\Big(h((F^\times)^d)\Big). \tag{4.2}$$

This is the group-theoretic center of $\widetilde{A}$, not the set of dth powers of elements of $\widetilde{A}$. The latter notation can be misleading for even n.

Lemma 4.1. *The formula (4.2) holds for every F containing μ_n. If $\gcd(p, n) = 1$, then*

$$[\widetilde{A} : Z_A] = d^2. \tag{4.3}$$

The commutator induces a nondegenerate alternating pairing on the finite quotient $\widetilde{A}/Z_A$.

Proof. The Hilbert symbol is nondegenerate on $F^\times/(F^\times)^n$. Its square is trivial against every b precisely when $a^2 \in (F^\times)^n$, equivalently when $a \in (F^\times)^d$. This proves the center assertion. When $p \nmid n$, the dth-power map is an automorphism of $1 + \varpi\mathcal{O}_F$, while $F^\times \simeq \varpi^{\mathbb{Z}} \times \mu_{q-1} \times (1 + \varpi\mathcal{O}_F)$ and $d \mid q - 1$. Consequently $|F^\times/(F^\times)^d| = d^2$. Nondegeneracy on the quotient is exactly the definition of its radical. □

Remark 4.2. The centralizer of the *whole* torus $\widetilde{A}$ in $\widetilde{G}$ is Z_A. For an individual regular element $a \in \widetilde{A}$, however, $C_{\widetilde{G}}(a)$ can be strictly larger than Z_A and strictly smaller than $\widetilde{A}$. This distinction matters in the definition of orbital integrals. In particular, one must not integrate over $\widetilde{A}\backslash\widetilde{G}$ unless a actually commutes with $\widetilde{A}$.

4.2 The Stone–von Neumann parametrization

Assume $\gcd(p, n) = 1$. Put

$$C = h(\varpi^{d\mathbb{Z}}\mathcal{O}_F^\times).$$

The preimage $\widetilde{C} \subset \widetilde{A}$ is a maximal abelian subgroup containing Z_A and $[\widetilde{A} : \widetilde{C}] = d$. Indeed the Hilbert symbol of two units is trivial in the tame case, and (4.1) makes $\widetilde{C}$ abelian. The index is d, which is the square root of the index in (4.3); maximality follows from the nondegeneracy of the commutator pairing.

Theorem 4.3 (Genuine representations of the covering torus)**.** *Let $\gcd(p, n) = 1$. For each smooth genuine character $\lambda : Z_A \to \mathbb{C}^\times$, choose an extension $\widetilde{\lambda}$ to $\widetilde{C}$. Then*

$$\chi_\lambda = \mathrm{Ind}_{\widetilde{C}}^{\widetilde{A}}\,\widetilde{\lambda} \tag{4.4}$$

is irreducible of dimension d. Its isomorphism class is independent of the extension, and $\lambda \mapsto \chi_\lambda$ is a bijection between genuine characters of Z_A and irreducible smooth genuine representations of $\widetilde{A}$. Their traces satisfy

$$\operatorname{tr} \chi_\lambda(a) = \begin{cases} d\lambda(a), & a \in Z_A, \\ 0, & a \in \widetilde{A} \setminus Z_A. \end{cases} \tag{4.5}$$

Proof. The quotient $\widetilde{A}/Z_A$ is a finite abelian group with a nondegenerate commutator pairing. Given a genuine central character λ, its extension to the maximal abelian $\widetilde{C}$ exists because $\widetilde{C}/Z_A$ is finite and $\mathbb{C}^\times$ is divisible. The stabilizer of an extension under conjugation by $\widetilde{A}$ is exactly $\widetilde{C}$, by nondegeneracy; Mackey's irreducibility criterion then gives an irreducible induction of degree $[\widetilde{A} : \widetilde{C}] = d$. Two extensions of the same λ are conjugate by $\widetilde{A}$, since the pairing identifies $\widetilde{A}/\widetilde{C}$ with the character group of $\widetilde{C}/Z_A$. Conversely, any smooth irreducible genuine representation has a genuine central character and is equivalent to this induction (the finite Stone–von Neumann theorem). For $a \in Z_A$, Schur's lemma gives $\chi_\lambda(a) = \lambda(a)\,\mathrm{id}$, hence the first case of (4.5). If $a \notin Z_A$, choose $b \in \widetilde{A}$ with $bab^{-1} = \zeta a$, $\zeta \neq 1$. Conjugation preserves trace, whereas genuineness multiplies it by ζ; thus the trace vanishes. □

Example 4.4 (The double and triple covers)**.** *For $n = 2$, $d = 1$. The covering torus is abelian, $Z_A = \widetilde{A}$, and genuine irreducible torus representations have dimension one. This does* not *imply that the double cover splits over A. For $n = 3$, $d = 3$ and $[\widetilde{A} : Z_A] = 9$. Each genuine torus representation is three-dimensional, induced from a maximal abelian subgroup of index three. Its trace is supported on Z_A and equals three times its central character there. These are the two smallest tests of every proposed formula for principal-series characters.*

Example 4.5 (Degrees four and six)**.** *For $n = 4$ one has $d = 2$ and $[\widetilde{A} : Z_A] = 4$. A genuine irreducible representation of $\widetilde{A}$ has dimension two, and the power-section $\nu(t) = (h(t^2), 1)$ is multiplicative since $(a^2, b^2)_4 = (a, b)_4^4 = 1$. Theorem 8.3 therefore applies in degree four.*

For $n = 6$ one has $d = 3$ and $[\widetilde{A} : Z_A] = 9$, but $(a^3, b^3)_6 = (a, b)_6^9$ need not be one. The torus character formula remains valid, whereas the same proposed power-section map need not be a homomorphism. This illustrates why the restriction on n in the transfer theorem is substantive.

Remark 4.6 (Correction of an earlier multiplicity claim). The two-to-one map from characters of $A_n = h(\varpi^{n\mathbb{Z}}\mathcal{O}_F^\times)$ to genuine torus representations asserted in the preceding version of this manuscript cannot hold uniformly. For odd n, $A_n = C$, so the displayed construction of two extensions collapses. The correct invariant is the genuine character of Z_A; this avoids artificial parity-dependent multiplicity claims.

5 Split orbital integrals and their image

5.1 The correct quotient

Let $C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ consist of genuine compactly supported locally constant test functions. Since Z_A centralizes all of $\widetilde{A}$, for every $a \in \widetilde{A}_{\mathrm{reg}}$ the expression

$$\mathcal{O}_f^{Z_A}(a) = |D(a)|^{1/2} \int_{Z_A \backslash \widetilde{G}} f(x^{-1}ax)\, d\dot{x} \tag{5.1}$$

is well-defined. Its quotient differs from the full centralizer of an individual a in general, but this causes no ambiguity because every element of Z_A centralizes a.

For $z \in (Z_A)_{\mathrm{reg}}$ the centralizer of z in $\widetilde{G}$ is $\widetilde{A}$. We therefore also set

$$\Phi_f(z) = |D(z)|^{1/2} \int_{\widetilde{A}\backslash\widetilde{G}} f(x^{-1}zx)\, d\dot{x}. \tag{5.2}$$

With quotient measures induced from the fixed Haar measures,

$$\mathcal{O}_f^{Z_A}(z) = d^2 \Phi_f(z) \qquad (z \in (Z_A)_{\mathrm{reg}}), \tag{5.3}$$

because $[\widetilde{A} : Z_A] = d^2$. This finite-index factor must not be confused with d, the dimension of an irreducible torus representation.

Lemma 5.1 (Vanishing)**.** *For $a \in \widetilde{A}_{\mathrm{reg}} \setminus Z_A$ one has $\mathcal{O}_f^{Z_A}(a) = 0$.*

Proof. Choose $b \in \widetilde{A}$ with $b^{-1}ab = \zeta a$ for some $\zeta \neq 1$. Since Z_A is normal in $\widetilde{A}$, left translation $x \mapsto bx$ preserves the measure on $Z_A\backslash\widetilde{G}$. The transformed integrand equals $f(x^{-1}b^{-1}abx) = \zeta^{-1} f(x^{-1}ax)$. Thus $\mathcal{O}_f^{Z_A}(a) = \zeta^{-1}\mathcal{O}_f^{Z_A}(a)$, proving the assertion. □

5.2 Iwasawa's formula and surjectivity

Here assume $\gcd(p, 2n) = 1$ and use the chosen splitting of K. For $g \in \widetilde{G}$ put

$$f^K(g) = \int_K f(k^{-1}gk)\, dk, \qquad \mathrm{vol}(K) = 1.$$

Using $\widetilde{G} = \widetilde{A}NK$ and the change of unipotent variable for $z \in (Z_A)_{\mathrm{reg}}$ gives

$$\Phi_f(z) = \delta_B(z)^{1/2} \int_N f^K(zu)\, du. \tag{5.4}$$

The right-hand side makes sense on all Z_A and is compactly supported and locally constant. We use it to define Φ_f at singular elements. On the regular set Φ_f is invariant under Weyl conjugation, hence its extension is also Weyl invariant.

Theorem 5.2 (Image of the split orbital-integral transform)**.** *If $\gcd(p, 2n) = 1$, the map*

$$C_c^\infty(\widetilde{G})_{\mathrm{gen}} \longrightarrow C_c^\infty(Z_A)_{\mathrm{gen}}^W, \qquad f \longmapsto \Phi_f \tag{5.5}$$

is surjective. Equivalently, the Z_A-quotient integral (5.1) restricts to the same space, with the factor d^2 from (5.3). The superscript W denotes invariance under conjugation by a lift of w, not naive inversion in $\widetilde{A}$.

Proof. Formula (5.4) proves that the map lands in the indicated space. For surjectivity, first take $\phi \in C_c^\infty((Z_A)_{\mathrm{reg}})_{\mathrm{gen}}^W$. On the open split regular locus, conjugation gives a finite Weyl covering of the corresponding conjugacy-saturated set; the usual submersion argument, with compactly supported local cutoffs along the conjugacy fibers, constructs f such that $\Phi_f = \phi$. The two Weyl charts give the same function precisely because ϕ is W-invariant.

For a general ϕ, only neighborhoods of the singular fibers remain. These fibers lie above I and $-I$. Choose a sufficiently small normal principal congruence subgroup $K_m \triangleleft K$ on which the cover is split. A genuine characteristic function of a central coset zK_m (extended equivariantly under μ_n) is invariant under K-conjugation. By (5.4), its transform is a nonzero constant on a sufficiently small neighborhood of z in Z_A and vanishes outside that neighborhood; rescaling gives a local characteristic function. The μ_n-equivariance and W-invariance are automatic for the resulting central fibers. Subtract a finite linear combination of these functions from ϕ; the remainder is compactly supported away from the singular set, where the first paragraph applies. This proves surjectivity. The same assertion is recorded, with a different quotient normalization, in [J4, Lemma 2.2]. □

Remark 5.3 (Satake and the image theorem). Equation (5.4) is a normalized constant-term expression for the *conjugation average* of a test function. It should not be confused without further argument with the spherical Satake isomorphism of convolution algebras. In particular, surjectivity of (5.5) for all test functions does not by itself prove a spherical Satake isomorphism.

6 Parabolic induction and known unitary representations

Let χ be a genuine irreducible smooth representation of $\widetilde{A}$, extended trivially to $\widetilde{B} = \widetilde{A}N$. The *normalized* principal series $\mathrm{I}(\chi) = \mathrm{Ind}_{\widetilde{B}}^{\widetilde{G}}(\chi \otimes \delta_B^{1/2})$ is realized on locally constant vector-valued functions satisfying

$$f(bg) = \delta_B(b)^{1/2}\chi(b)f(g), \qquad b \in \widetilde{B},\ g \in \widetilde{G}. \tag{6.1}$$

The action is by right translation. This is often called *normalized parabolic induction*; it is unitary if χ is unitary, but it need not be unitary for nonunitary χ.

6.1 Jacquet modules and ordinary principal series

The geometric lemma and the Jacquet functor carry over to finite central extensions split over the unipotent radical [BZ1, BZ2, KP]. In rank one, the semisimplification of the normalized Jacquet module of $\mathrm{I}(\chi)$ has the two terms χ and $\chi^{\tilde{w}}$, with

$$\chi^{\tilde{w}}(a) = \chi(\tilde{w}^{-1}a\tilde{w}). \tag{6.2}$$

The twist in (6.2) is *not* generally $\chi(a^{-1})$: the inverse of a lift includes its cocycle and its central component behaves differently under inversion.

Proposition 6.1. *If χ is unitary, $\mathrm{I}(\chi)$ is a unitary, tempered representation (possibly reducible). Its irreducible constituents are unitary and tempered. If χ is Weyl-regular, $\chi^{\tilde{w}} \not\simeq \chi$, then $\mathrm{I}(\chi)$ is irreducible in the standard rank-one unitary principal-series setting; in particular this assertion holds under Moen's tame hypotheses.*

Proof. The compact picture of normalized induction supplies a positive $\widetilde{G}$-invariant Hermitian form when χ is unitary; temperedness follows from tempered parabolic induction. For the stated irreducibility criterion in the tame rank-one setting, see [Mo2] and the standard geometric-lemma/intertwining argument. No extension of χ to a torus of a cover of $\mathrm{GL}(2, F)$ is needed or asserted here. □

Remark 6.2 (Reducibility). In the tame cases studied by Moen, the exceptional reducible unitary principal series depend on parity and self-association of the inducing data [Mo2]. These constituents must be included in any complete unitary-dual account. The present note does not restate a universal "two constituents for every Weyl-fixed character" assertion; Goldberg–Szpruch [GoldSz] gives a useful later analysis of Plancherel measures and reducibility.

6.2 Complementary series and endpoints: literature and limits

A *complementary-series representation* is a non-tempered unitarizable irreducible normalized principal series whose inducing representation is not unitary. Fix a unitary genuine central character λ_0 of Z_A, and consider suitable Weyl-self-dual families χ_{λ_s} with unramified real parameter s. An invariant positive form, when it exists, is constructed from a suitably normalized standard intertwining operator

$$M(s) : \mathrm{I}(\chi_{\lambda_s}) \longrightarrow \mathrm{I}(\chi^{\tilde{w}}_{\lambda_{-s}}).$$

Its positivity is a substantive assertion, not a consequence of formal uniqueness of invariant Hermitian forms. One must also establish the correct self-duality, nonzero intertwining operator, normalization and absence of reducibility in the claimed interval.

For the genuine double cover in odd residual characteristic, the classical Weil-representation and intertwining theory treats the complementary-series behavior and the endpoint at $s = 1/2$ [G, GS]. Aritürk treats the analogous threefold problem with $p > 3$ and parameter $s = 1/3$ at the endpoint [Ar]. The familiar square-integrable (special) constituent and the non-tempered constituent at these endpoints should be understood *within the hypotheses of the cited sources*. In particular, it is not justified to assert from the $\mathrm{GL}(2, F)$ literature alone that for every n, including $p \mid n$, the whole interval $0 < s < 1/n$ is unitarizable or that the endpoint has the same composition series.

6.3 Why the intertwining-operator argument needs care

The earlier draft introduced a family of intertwiners by

$$(T_s\varphi)(g, a) = \int_F \varphi\Big(\tilde{w}\, u(x) g, \tilde{w} a \tilde{w}^{-1}\Big)\, dx,$$

initially in a region of absolute convergence, and represented its Fourier-transformed action by a matrix J_s of size $d \times d$. This is a useful computational approach and closely follows Aritürk's treatment of the cubic cover [Ar]. However, three logically distinct steps are involved:

(i) meromorphic continuation and the correct normalization of T_s, including the Weyl-conjugate inducing datum;

(ii) construction of a nondegenerate invariant Hermitian form from T_s, for a family that is genuinely Hermitian self-dual;

(iii) positivity of that form, equivalently positivity of the relevant normalized operator or matrix J_s, throughout the proposed interval.

None of these implications follows merely because an analogous representation of a cover of $\mathrm{GL}(2,F)$ is known: extension of an irreducible $\mathrm{SL}(2,F)$ representation to the larger cover must first be proved. Nor does Schur's lemma alone establish positivity; it only controls the scalar ambiguity between forms when the representation is already irreducible. In higher covering degree, J_s is matrix-valued, making the positivity step especially significant. The results of [GoldSz] provide additional tools for analyzing the Plancherel measure and reducibility.

The original integral and its matrix interpretation are therefore retained here as motivation, but without converting them into an unproved arbitrary-degree complementary-series theorem.

6.4 Supercuspidal representations and completeness

An irreducible admissible genuine representation is supercuspidal if its Jacquet module for N vanishes, equivalently if its matrix coefficients are compactly supported modulo the center. The general rank-one subquotient theorem says that every irreducible admissible genuine representation is either supercuspidal or a subquotient of some $\mathrm{I}(\chi)$. The problem of identifying the *unitary* subquotients is additional. Manderscheid's work [M1, M2] treats the double cover; the author's earlier contributions [J2, J3] concern specified supercuspidal correspondences and Hecke-algebra contexts. These citations are not, by themselves, a proof of an exhaustive unitary-dual theorem for arbitrary n and wild residual characteristic.

7 Principal-series characters

Theorem 7.1 (Corrected principal-series character)**.** *Assume* $\gcd(p,n)=1$. *Let* λ *be a genuine smooth central character of* Z_A, $\chi=\chi_\lambda$ *as in Theorem 4.3, and* $\Pi=\mathrm{I}(\chi)$, *not necessarily irreducible. Its distribution character is represented on the regular semisimple set by a locally integrable function whose values on the split part are*

$$\Theta_\Pi(a)=\frac{\operatorname{tr}\chi(a)+\operatorname{tr}\chi^{\tilde w}(a)}{|D(a)|^{1/2}}\qquad (a\in \tilde A_{\mathrm{reg}}), \tag{7.1}$$

and which vanishes on elliptic regular semisimple classes. In particular, the split regular formula is zero off Z_A *and, for* $z\in (Z_A)_{\mathrm{reg}}$, *is*

$$\Theta_\Pi(z)=\frac{d\big(\lambda(z)+\lambda(\tilde w^{-1}z\tilde w)\big)}{|D(z)|^{1/2}}. \tag{7.2}$$

Both formulas have the correct transformation law $\Theta_\Pi(\zeta a)=\zeta\Theta_\Pi(a)$.

Proof. The rank-one induced-character formula is obtained by restricting the character of normalized parabolic induction to the regular semisimple locus. A split regular element has

two fixed points on G/B, represented by the two Weyl elements; their contributions are the inducing character evaluated at a and at the Weyl *conjugate* of a, divided by the square root of the Weyl discriminant. An elliptic regular element has no F-rational fixed point on G/B and contributes zero. The proof carries over to the finite central extension because it splits over N; compare the induced-character arguments in [G, F, FK]. Formula (7.2) follows from (4.5). The central-equivariance claim is immediate because both $\chi(\zeta a)$ and $\chi^{\tilde{w}}(\zeta a)$ are ζ times their respective values at a. □

Caution 7.2 (Why inversion cannot replace Weyl conjugation). For $n > 2$, the expression $\operatorname{tr}\chi(a) + \operatorname{tr}\chi(a^{-1})$ transforms under $a \mapsto \zeta a$ with factors ζ and ζ^{-1}, respectively, so its quotient by $|D(a)|^{1/2}$ is generally not a genuine representation character. In particular it cannot be used to prove a matching theorem. The correction in (7.1) is essential even when a particular cocycle section makes Weyl conjugation look like ordinary inversion on the underlying *matrix*.

Corollary 7.3 (Trace in terms of normalized orbital integrals)**.** *Assume* $\gcd(p, 2n) = 1$ *and the measures of Section 3. For* $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$,

$$\operatorname{tr}\Pi(f) = \frac{1}{2}\int_{(Z_A)_{\mathrm{reg}}} \Big(\operatorname{tr}\chi(z) + \operatorname{tr}\chi^{\tilde{w}}(z)\Big)\Phi_f(z)\,dz \tag{7.3}$$

$$= d\int_{Z_A} \lambda(z)\Phi_f(z)\,dz. \tag{7.4}$$

The integral on the second line uses the locally constant extension of Φ_f *and the singular set has Haar measure zero.*

Proof. Apply the Weyl integration formula to the ordinary central-invariant product $\Theta_\Pi f$. Since the character vanishes on elliptic regular elements and on split regular $\widetilde{A} \setminus Z_A$, only Z_A contributes; its elements have centralizer $\widetilde{A}$. The Weyl factor is $1/2$, giving the first identity by (7.1) and (5.2). The second follows from (4.5) and Weyl invariance of Φ_f. Note that the genuine λ cancels the inverse central character of Φ_f, so the integrand is well-defined on Z_A. □

8 A restricted and normalized split-torus transfer

In this section assume $\gcd(p, 2n) = 1$ *and*

$$n \text{ is odd} \qquad \text{or} \qquad 4 \mid n. \tag{8.1}$$

These restrictions imply $n \mid d^2$. For $t \in F^\times$ define

$$\nu(h(t)) = (h(t^d), 1) \in Z_A. \tag{8.2}$$

Lemma 8.1. *Under* (8.1), $\nu : A \to Z_A$ *is a continuous homomorphism with kernel of order* d. *Its image is* $Z_A^0 = \{(h(u), 1) : u \in (F^\times)^d\}$, *and* $Z_A = \mu_n\, Z_A^0$ *as a direct product. The map is Weyl-equivariant:*

$$\tilde{w}^{-1}\nu(h(t))\tilde{w} = \nu(h(t^{-1})). \tag{8.3}$$

If dt is the chosen Haar measure on A and dz its restriction to Z_A^0, then

$$\int_{Z_A^0} H(z)\,dz = \frac{1}{d}\int_A H(\nu(t))\,dt \tag{8.4}$$

for every compactly supported H on Z_A^0.

Proof. On the diagonal torus, (3.1) gives $\beta(h(a),h(b)) = (b,a)_n$. On dth powers this equals $(b,a)_n^{d^2} = 1$ by (8.1); hence ν is multiplicative and its image splits the central extension above $h((F^\times)^d)$. The kernel consists of dth roots of unity, of which there are d since $d \mid n$ and $\mu_n \subset F$. A direct substitution in (3.1) shows $\tilde{w}^{-1}(h(u),1)\tilde{w} = (h(u^{-1}),1)$; one uses $(u,u)_n^2 = 1$. This proves (8.3).

For the measure assertion, normalize dt temporarily by $\mathrm{vol}(\mathcal{O}_F^\times) = 1$. In the tame case the subgroup $(\mathcal{O}_F^\times)^d$ has index d, and the dth-power map sends $\mathcal{O}_F^\times$ onto it. Thus the pushforward of dt has d times the Haar measure restricted from A. The same factor holds on each valuation coset, proving (8.4). □

Let F_f^A be the ordinary normalized split orbital integral, with the same Haar convention and its locally constant extension over A,

$$F_f^A(t) = |D(t)|^{1/2}\int_{A\backslash G} f(x^{-1}tx)\,d\dot{x} \quad (t \in A_{\mathrm{reg}}).$$

Definition 8.2 (Matching on the split torus)**.** *For $\bar{f} \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ and $f \in C_c^\infty(G)$ we say that they are* split-matching *if*

$$F_f^A(t) = \Phi_{\bar{f}}(\nu(t)) \qquad (t \in A), \tag{8.5}$$

where both sides use their locally constant extensions at the singular points. This notion concerns split orbital integrals only.

Theorem 8.3 (Restricted principal-series transfer)**.** *Assume* $\gcd(p,2n) = 1$ *and* (8.1)*. For every genuine irreducible χ_λ of $\widetilde{A}$, let*

$$\eta_\lambda(t) = \lambda(\nu(t)), \qquad t \in A, \tag{8.6}$$

which is an ordinary smooth character of A. If $\bar{f}$ and f are split-matching, then

$$\boxed{\mathrm{tr}\,\mathrm{I}(\chi_\lambda)(\bar{f}) = \mathrm{tr}\,\mathrm{Ind}_B^G(\eta_\lambda\,\delta_B^{1/2})(f).} \tag{8.7}$$

Every $\bar{f} \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ has at least one split-matching $f \in C_c^\infty(G)$. If λ is unitary then η_λ is unitary. No assertion about irreducibility of the two induced representations is needed.

Proof. By Lemma 8.1, η_λ is a character, and (8.3) identifies the two Weyl terms in the ordinary principal-series character formula. For a genuine integrand $\lambda(z)\Phi_{\bar{f}}(z)$, integration

over $Z_A = \mu_n Z_A^0$ is the same as integration over Z_A^0; the central factors cancel. Using Corollary 7.3 and (8.4) gives

$$\begin{aligned}\operatorname{tr} \mathrm{I}(\chi_\lambda)(\bar{f}) &= d \int_{Z_A} \lambda(z)\Phi_{\bar{f}}(z)\,dz \\ &= d \int_{Z_A^0} \lambda(z)\Phi_{\bar{f}}(z)\,dz \\ &= \int_A \eta_\lambda(t)\Phi_{\bar{f}}(\nu(t))\,dt \\ &= \int_A \eta_\lambda(t) F_f^A(t)\,dt.\end{aligned}$$

The ordinary rank-one induced-character formula, together with Weyl integration, gives

$$\operatorname{tr} \operatorname{Ind}_B^G(\eta_\lambda \delta_B^{1/2})(f) = \int_A \eta_\lambda(t) F_f^A(t)\,dt,$$

since F_f^A is Weyl invariant. This proves the equality.

For existence, Theorem 5.2 shows that $\Phi_{\bar{f}} \in C_c^\infty(Z_A)^W_{\mathrm{gen}}$. The pullback $\Phi_{\bar{f}} \circ \nu$ is compactly supported, locally constant and Weyl invariant on A. The usual (linear) $\mathrm{SL}(2, F)$ split orbital-integral image theorem, the $n = 1$ instance of Theorem 5.2, produces a test function f whose transform is that pullback. Finally unitary characters remain unitary upon pullback. □

Example 8.4 (Checking the normalization in degree three)**.** *Suppose $n = 3$, $p \neq 2, 3$, and λ is a genuine central character trivial on the chosen split units. Let $\bar{f}_0$ be the genuine idempotent of the split maximal compact K, normalized so that $\mathrm{vol}(K) = 1$. On Z_A, its normalized constant term is one on the unit subgroup (with the required central equivariance) and zero off it. Since $[\mathcal{O}_F^\times : (\mathcal{O}_F^\times)^3] = 3$, we obtain*

$$\operatorname{tr} \mathrm{I}(\chi_\lambda)(\bar{f}_0) = 3 \int_{Z_A} \lambda(z)\Phi_{\bar{f}_0}(z)\,dz = 3 \cdot \frac{1}{3} = 1.$$

This agrees with the expected dimension of the spherical fixed space and explains the factor d in (7.4). It also shows why replacing the subgroup index d^2 by d inside (5.3) would be incorrect.

Remark 8.5 (The omitted parity case). If $n \equiv 2 \pmod 4$, then $n \nmid d^2$, so the map $t \mapsto (h(t^d), 1)$ is generally *not* multiplicative. A bare phrase such as "twist by the Weil factor" does not define a μ_n-valued group homomorphism in the present cover. Weil indices may naturally take values in μ_4 and require a compatible normalization or an enlarged central group. The comparison (8.7) is therefore *not claimed here* for $n \equiv 2 \pmod 4$ (including $n = 2$). The double-cover correspondence is meaningful and well studied, but must be written using its own correctly normalized genuine character and transfer factor [G, GS, F, Sch].

Remark 8.6 (Spherical restriction). With a fixed splitting of K, the identity (8.7) also applies *whenever* $\bar{f}$ and f happen to be spherical and satisfy (8.5). The existence of a spherical partner for every spherical $\bar{f}$ is a separate assertion about metaplectic Satake transforms; it is not a consequence of the smooth matching theorem above and is not asserted without an additional argument.

9 Conclusions and limits of the correction

The structural content established in this version is the parametrization and character theory of the tame covering torus, the corrected image theorem for split orbital integrals, the Weyl-conjugate principal-series character formula, and the restricted split character comparison of Theorem 8.3. The calculations distinguish three integers that should not be confused:

$$\dim \chi_\lambda = d, \qquad [\widetilde{A} : Z_A] = d^2, \qquad |\ker \nu| = d \quad \text{(in the allowed transfer degrees)}.$$

The following are not consequences of the present proofs: (i) a full description of the unitary dual for arbitrary covering degree and wild residue characteristic; (ii) a uniform complementary-series theorem with endpoint $1/n$; (iii) spherical matching for arbitrary spherical test functions; (iv) the power-map transfer in degree $n \equiv 2 \pmod 4$; (v) elliptic orbital-integral or full Jacquet–Langlands matching. Each is a possible subject for additional research, rather than a result silently inherited from the 1999 preprint.

10 Glossary

Genuine A representation with central action $\zeta \mapsto \zeta$; a trace test function transforms by ζ^{-1}.

Covering-torus radical The group-theoretic center of $\widetilde{A}$; it is the radical of the Hilbert-symbol commutator pairing.

Weyl conjugate The representation $\chi^{\tilde{w}}(a) = \chi(\tilde{w}^{-1} a \tilde{w})$, which must be distinguished from composition with group inversion.

Normalized orbital integral An orbital integral multiplied by $|D|^{1/2}$; its quotient group must centralize the element.

Stone–von Neumann theory Parametrization of irreducible genuine representations of a two-step nilpotent covering torus by their central characters.

Complementary series Non-tempered, unitarizable induced representations obtained for certain real, nonunitary parameters.

Supercuspidal An irreducible admissible representation with vanishing proper Jacquet modules.

Spherical Invariant under the image of a chosen splitting $K \hookrightarrow \widetilde{G}$; existence of the splitting is an extra hypothesis.

Split-matching Equality (8.5) of normalized split torus orbital integrals, not a full endoscopic or stable orbital integral transfer.

Acknowledgments and revision history

The 2026 revision began with an Argon-Gemini manuscript and an independent referee report by ChatGPT (OpenAI). A subsequent attempt to revise the manuscript using Argon-Gemini was not completed; the reason for that difficulty is not established. ChatGPT prepared a corrected LaTeX edition with revised proofs, examples, literature context, and explicit limits on assertions not justified by the preceding version. The present editorial revision reorganizes the notation, adds a separate literature review, and revises the typesetting of the notation table. Computational and typesetting checks do not substitute for independent mathematical refereeing.

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References

[Ar] H. Aritürk, "On the composition series of principal series representations of a three-fold covering group of $SL(2, K)$", *Nagoya Math. J.* **77** (1980), 177–196.

[BZ1] I. N. Bernstein and A. V. Zelevinsky, "Representations of the group $GL(n, F)$, where F is a non-archimedean local field", *Russian Math. Surveys* **31** (1976), 1–68.

[BZ2] I. N. Bernstein and A. V. Zelevinsky, "Induced representations of reductive p-adic groups. I", *Ann. Sci. Èc. Norm. Sup.* **10** (1977), 441–472.

[F] Y. Flicker, "Automorphic forms on covering groups of $GL(2)$", *Invent. Math.* **57** (1980), 119–182.

[FK] Y. Flicker and D. Kazhdan, "Metaplectic correspondence", *Publ. Math. IHÉS* **64** (1986), 53–110.

[G] S. Gelbart, *Weil's Representation and the Spectrum of the Metaplectic Group*, Lecture Notes in Mathematics 530, Springer, 1976.

[GS] S. Gelbart and P. Sally, "Intertwining operators and automorphic forms on the metaplectic group", *Proc. Natl. Acad. Sci. USA* **72** (1975), 1406–1410.

[GoldSz] D. Goldberg and D. Szpruch, "Plancherel measures for coverings of p-adic $SL_2(F)$", *Int. J. Number Theory* **12** (2016), 1907–1936.

[J1] D. Joyner, "On the metaplectic analog of Kazhdan's "endoscopic" lifting", *Israel J. Math.* **61** (1988), 113–154.

[J2] D. Joyner, "A correspondence for the generalized Hecke algebra of the metaplectic cover $\overline{SL(2, F)}$, F p-adic", *New York J. Math.* **4** (1998), 223–235.

[J3] D. Joyner, "A correspondence for the supercuspidal representations of $\overline{SL_2(F)}$, F p-adic", *Arch. Math.* **73** (1999), 332–340.

[J4] D. Joyner, "Invariant distributions on the n-fold metaplectic covers of $GL(r, F)$, F p-adic", *J. Fourier Anal. Appl.* **7** (2001), no. 4, 343–358.

[KP] D. Kazhdan and S. Patterson, "Metaplectic forms", *Publ. Math. IHÉS* **59** (1984), 35–142.

[K] T. Kubota, "Topological covering of $SL(2)$ over a local field", *J. Math. Soc. Japan* **19** (1967), 114–121.

[M1] D. Manderscheid, "On the supercuspidal representations of $SL(2)$ and its 2-fold cover. I", *Math. Ann.* **266** (1987), 287–295.

[M2] D. Manderscheid, "On the supercuspidal representations of $SL(2)$ and its 2-fold cover. II", *Math. Ann.* **266** (1987), 297–305.

[Mo1] C. Moen, *Intertwining Operators for Covering Groups of $SL(2)$ over a Local Field*, Ph.D. thesis, University of Chicago, 1979.

[Mo2] C. Moen, "Irreducibility of unitary principal series representations for covering groups of $SL(2)$", *Pacific J. Math.* **135** (1988), 89–110.

[Sch] J. Schultz, *Lifting of Characters of $\overline{SL_2(F)}$ and $SO_{2,1}(F)$, for F Nonarchimedean*, Ph.D. thesis, University of Maryland, 1998.

[Szpruch19] D. Szpruch, "On Shahidi local coefficients matrix, *Manuscripta Math.* extbf159 (2019), nos. 1–2, 117–159.

[Tang] S. Tang, "Principal series representations of metaplectic groups", arXiv:1706.05145 (2017).